Tutorial Sheet – ArrayLists (Revision)

Programming Fundamentals 1
Topic: Introduction to ArrayLists
Purpose: Revision before moving on to advanced Java topics

Part A: Understanding ArrayLists

1. Declaring and Creating ArrayLists

Write Java code for each of the following:

1. Declare and create an `ArrayList<String>` called `names`.
2. Declare and create an `ArrayList<Integer>` called `scores`.
3. Add three names to the `names` ArrayList.
4. Add three integer values to the `scores` ArrayList.

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2. Predicting Output

Without running the code, write down exactly what is printed:

```
ArrayList<Integer> numbers = new ArrayList<Integer>();
numbers.add(5);
numbers.add(10);
numbers.add(15);

System.out.println(numbers.get(1));
System.out.println(numbers.size());
```

Answer the following:

a. What is printed on the first line?

b. What is printed on the second line?

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3. Understanding Indexing

Consider the following code:

```
ArrayList<String> colours = new ArrayList<String>();
colours.add("Red");
```

```java
colours.add("Green");
colours.add("Blue");
```

Answer the following:

a. What element is stored at index `0`?

b. What element is stored at index `2`?

c. What happens if `colours.get(3)` is executed?

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4. Correcting Logic

The following code is intended to print all elements in an ArrayList, but it contains an error.

```java
ArrayList<String> animals = new ArrayList<String>();
animals.add("Dog");
animals.add("Cat");
animals.add("Horse");

for (int i = 0; i <= animals.size(); i++) {
    System.out.println(animals.get(i));
}
```

a. Explain what is wrong with the loop condition.

b. Correct the code.

Part B: Using Loops with ArrayLists

5. Looping with a for Loop

Write Java code to:

1. Print all elements in the `names` ArrayList using a `for` loop.
2. Print all values in the `scores` ArrayList using a `for` loop.

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6. Looping with a for-each Loop

Given the following ArrayList:

```java
ArrayList<String> modules = new ArrayList<String>();
modules.add("Programming");
```

```
modules.add("Databases");
modules.add("Web Development");
```

1. Write a for-each loop to print all module names.
2. Explain one limitation of using a for-each loop with an ArrayList.

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7. Removing Elements

Consider the following code:

```
ArrayList<Integer> values = new ArrayList<Integer>();
values.add(10);
values.add(20);
values.add(30);
values.remove(1);
```

Answer the following:

a. Which value is removed from the list?

b. What value is now stored at index `1`?

c. What is the size of the ArrayList after removal?

Part C: Applying ArrayLists

8. Checking for Empty Lists

Write Java code to check if an `ArrayList<Product>` called `products` is empty. Use the most appropriate method.

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9. Finding the Lowest Value

You are given the following ArrayList containing **five integer values**:

```
prices = [25, 40, 15, 60, 30]
```

(Assume the values are stored in an `ArrayList<Integer>` in the order shown.)

Part A – Pseudocode

Write pseudocode that describes the steps needed to find the **lowest value** in the ArrayList.

Your pseudocode should: - Make an initial assumption - Compare values one at a time - Update the lowest value when needed

Part B – Java Code

Using your pseudocode as a guide, write Java code to find and print the lowest value in the `prices` ArrayList.

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10. Thinking Challenge

Answer the following:

1. Give one advantage of using an ArrayList instead of an array in a program such as a `Store` class.
2. Why is it no longer necessary to keep a separate `total` variable when using an ArrayList?

End of Tutorial Sheet

Attempt all questions before checking the solutions.